

Experimental Aircraft

These were exciting times, as European nations and the US experimented with supersonic flight, delta-wing layouts, extreme wing sweeps, vertical takeoff and landing (VTOL), and alternative power from ramjets and rockets. Some sensational prototypes were built and many valuable lessons learned. However, the life of a test pilot in this world was fraught with danger: many lost their lives in testing accidents.



△ Shorts SB5 1952

Origin UK

Engine 4,850 lb (2,200 kg) thrust Bristol BE26 Orpheus turbojet (earlier, 3,500 lb /1,588 kg thrust Rolls-Royce Derwent 8)

Top speed 403 mph (650 km/h)

Built to check ideal wing angles and tailplane positions for low-speed handling of the proposed Lightning fighter, the SB5's wooden wings could be set at 50 degrees, 60 degrees, or 69 degrees, the greatest sweep yet tried.

▽ Shorts SC1 1957

Origin UK

Engine 5 x 2,130 lb (966 kg) thrust Rolls-Royce RB108 turbojet

Top speed 246 mph (396 km/h)

The first British vertical takeoff aircraft with fly-by-wire controls, the SC1 had four engines for vertical operation and a fifth for forward propulsion. Nose, tail, and wingtip bleeds from the four lift engines gave it low-speed stability.



△ Sud-Ouest SO9000-01 Trident 1953

Origin France

Engine 2 x 1,654 lb (750 kg) thrust MD 30 Viper ASV.5 turbojet + 8,325 lb (3,776 kg) thrust SEPR 481 3-chamber liquid fuel rocket

Top speed 1,060 mph (1,706 km/h)

Development began in 1948 for France's supersonic interceptor, using wingtip-mounted turbojets plus a three-chamber rocket; 10 preproduction aircraft followed the two prototypes, but no more.

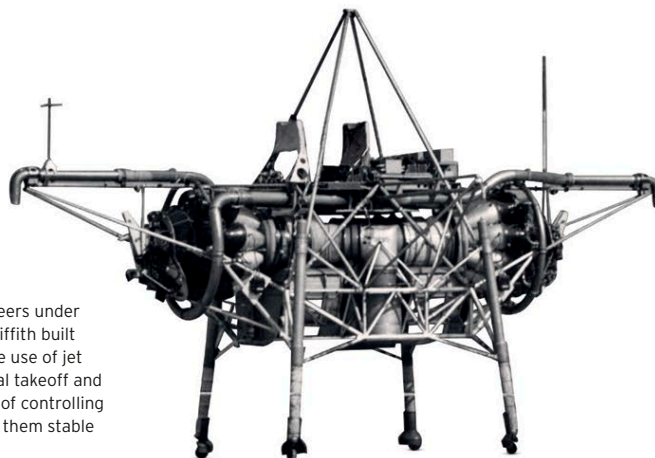
▷ Rolls-Royce Thrust Measuring Rig "Flying Bedstead" 1953

Origin UK

Engine 2 x 4,050 lb (1,837 kg) thrust Rolls-Royce Nene turbojet

Top speed N/A

Rolls-Royce engineers under Dr. Alan Arnold Griffith built two rigs to test the use of jet engines for vertical takeoff and to develop means of controlling them and keeping them stable when hovering.



△ Boulton Paul P.111 1950

Origin UK

Engine 5,100 lb (2,313 kg) thrust Rolls-Royce Nene R3N2 turbojet

Top speed 649 mph (1,045 km/h)

Designed by Dr. S. C. Redshaw for the UK Air Ministry, to test tail-less delta-wing characteristics, the P.111 had fully powered controls and tested a selection of fiberglass wing extensions.



▷ Lockheed XFV-1 1954

Origin US

Engine 5,332 hp Allison XT40-A-14 double-turboprop

Top speed 580 mph (933 km/h)

The US Navy requested a vertical takeoff aircraft to operate from small platforms on normal ships. The XFV-1, known as the "Pogo" actually flew vertically and did transition to level flight in tests, but it was too slow.

